WRITING SAMPLE DISCLAIMER: This is a hypothetical demonstration document prepared by Nathan Smith, adapting the governance framework, pillar structure, and terminology from his 2024 doctoral dissertation (Colorado Technical University, D.C.S. in Big Data Analytics) into a practical, employer-agnostic documentation workflow. The employer name, program name, prompt text, and AI-generated content shown below are entirely fictional and created solely to illustrate method. No classified, export-controlled, or proprietary information is included.

DOC-NBS-SAMPLE-06 · GUIDANCE DOCUMENT · ISSUE 1

AI-Assisted Technical Documentation: A Governance Workflow

Applying the LLM Usage Governance Framework to an AI-assisted work instruction, end to end

PREPARED BY	SOURCE FRAMEWORK	APPLICABILITY	STATUS
Nathan Smith	LLM Usage Governance Framework (Smith, 2024)	AI-assisted procedures, work instructions, guidance docs	Hypothetical / Demonstration

01 PURPOSE & BACKGROUND

My 2024 doctoral dissertation, *Governance Frameworks for Large Language Models in Technical Documentation to Support Small Defense Contracting Firms*, developed and evaluated the **LLM Usage Governance Framework** through a literature review and semi-structured interviews with ten technical communicators working in defense contracting. The framework organizes responsible AI use in technical documentation around five components: Ethical Guidelines, Data Security Protocols, Quality Assurance Processes, Training and Education, and Governance Structure.

This sample translates that research artifact into something a documentation team can actually run: a short procedure for taking a Copilot-assisted draft from raw SME input to a signed-off, compliant work instruction, with the review gates the framework calls for built directly into the steps rather than left as an abstract policy statement.

02 FRAMEWORK OVERVIEW

The five components below carry through into the workflow in Section 3 as concrete checkpoints, not just principles.

COMPONENT	WHAT IT GOVERNS
Ethical Guidelines	Human review of anything AI drafts, disclosure of where AI was used, and active checks for bias or misleading phrasing before content reaches a technician or operator.
Data Security Protocols	What may and may not be entered into an AI tool — classification review, export-control screening, and access controls on any AI-generated output before it's stored or distributed.
Quality Assurance Processes	Verification of AI-drafted content against released engineering data, a defined edit pass for standards compliance (e.g., STE, S1000D, style guide), and a documented review record.
Training & Education	Building prompt-engineering competence and AI-literacy across the writing team so drafts arrive closer to final and reviewers know what to look for.
Governance Structure	Named roles and a decision path for approving, escalating, or rejecting AI-assisted content, plus an audit trail of what was generated, by whom, and how it was reviewed.

03 WORKED EXAMPLE: SME NOTE → AI DRAFT → REVIEWED WORK INSTRUCTION

The scenario below is fictional: a hypothetical fuel-filter inspection task, used only to illustrate the workflow.

STEP 1 · RAW SME INPUT

```
Engineer note (as received): "tech needs to check the fuel filter differential
pressure gauge before every 50 hr inspection, if its in the yellow they should still be ok but if its
red they need to swap the filter element right away and if they're not sure they should probably just
swap it anyway to be safe since its cheap insurance"
```

STEP 2 · PROMPT ENGINEERING (ETHICAL GUIDELINES + TRAINING PILLARS)

A disclosed, logged prompt — not a one-off ad hoc request — run through the team's approved AI tool:

```
Rewrite the following maintenance note as a numbered procedure in Simplified
Technical English (ASD-STE100): one instruction per step, imperative mood, no idioms or vague
qualifiers. Preserve every decision condition exactly as given. Do not invent tolerances, part
numbers, or intervals that are not present in the source note. Flag anything ambiguous instead
of guessing.

Source note: [pasted verbatim above]
```

STEP 3 · RAW AI OUTPUT (UNREVIEWED)

```
1. Check the fuel filter differential pressure gauge before every 50-hour inspection.
2. If the indicator is in the yellow range, continue operation.
3. If the indicator is in the red range, replace the filter element immediately.
4. FLAGGED BY AI: source note does not define the yellow/red boundary values, and suggests
   replacing "anyway" if uncertain -- this is a judgment call, not a written tolerance. Confirm
   intended guidance before publishing.
```

WHY THIS STEP MATTERS: The AI correctly held the line rather than inventing a pressure value it wasn't given — and correctly flagged that "swap it anyway to be safe" is operator guidance, not a specification. Rewarding that behavior in the prompt (rather than asking the model to "just clean it up") is what keeps AI output inside the Ethical Guidelines and Data Security pillars: no fabricated tolerances entering a work instruction.

STEP 4 · HUMAN QA REVIEW (QUALITY ASSURANCE PILLAR)

- ☐ Verified against released engineering data: differential pressure setpoints confirmed against the current Package Data Sheet, not left to the AI's flagged assumption.
- ☐ Checked for fabricated content: no part numbers, tolerances, or intervals invented by the model; the one flagged gap was resolved by the SME, not guessed.
- ☐ STE / style compliance pass completed: imperative mood, one action per step, controlled vocabulary.
- ☐ No classified, export-controlled, or supplier-proprietary data was entered into the AI tool at any point in this workflow (Data Security Protocols).
- ☐ Reviewer initials and date logged in the audit record (Section 4).

STEP 5 · FINAL, REVIEWED WORK INSTRUCTION

```
1. Check the fuel filter differential pressure indicator before every 50-hour inspection.
2. If the indicator reads below 15 psid (yellow band), continue normal operation.
3. If the indicator reads 15 psid or above (red band), replace the filter element before
   continuing operation.

NOTE
Do not delay filter replacement once the indicator reaches the red band. Early replacement
at operator discretion is acceptable and does not require additional approval.
```

04 GOVERNANCE STRUCTURE & AUDIT RECORD

Every AI-assisted document carries a short record of who generated what and who checked it — the same accountability chain any SME-authored content already goes through, extended to cover the AI step.

FIELD	ENTRY
AI tool & version	e.g., Microsoft Copilot (Enterprise), model/version as configured by IT policy
Prompt used	Logged verbatim (see Step 2) for traceability and reuse
Source data authored by	SME / Engineer name, date
AI draft reviewed by	Technical Writer name, date
Technical accuracy verified by	SME sign-off against released engineering data, date
QA / compliance check by	Documentation Lead, date
Disposition	☐ Approved ☐ Returned for revision ☐ Rejected

Governance note: The point of the audit record isn't paperwork for its own sake — it's what lets a documentation lead answer, six months later, exactly which steps in a published manual came from an AI draft, what prompt produced them, and who verified them against source data. That traceability is what turns "we use AI to help write manuals" into something a quality audit can actually pass.

Research Study [Doctoral dissertation, Colorado Technical University]. The scenario, prompt, AI output, and part/parameter values in Sections 3–4 are fictional and constructed solely for this writing sample.